From a single characterisation snapshot to end-of-life: a PyBaMM-trained neural operator for second-life LFP RUL prediction

Krishna Chaitanya Vaddepally[a,*]

[a] *Turno (Blubble Pvt. Ltd.), Bengaluru, India*

Abstract

Repurposing used lithium-iron-phosphate (LFP) cells for stationary battery energy storage systems (BESS) is bottlenecked by the cycling budget required to calibrate per-cell degradation models: reliable extrapolation to end-of-life with the PyBaMM reaction-limited SEI ODE requires at least ∼400 cycles of measured state-of-health (SoH) trajectory per cell. Prior work has improved the quality of that per-cell calibration but has not shrunk the cycling budget itself. We propose a two-stage pipeline that decouples cycling budget from extrapolation horizon. First, PyBaMM's `Prada2013` chemistry with beginning-of-life parameters identified from OCV, GITT, HPPC, and RPT characterisation of a fresh cohort is used to generate a corpus of 300 synthetic degradation trajectories via a Sobol sweep across five degradation channels (SEI kinetics, plating, positive- and negative-electrode loss of active material, and SEI molar volume). Second, a DeepONet-style neural operator is trained on this corpus, mapping a five-stream input — DCIR, RPT capacity and IC-peak features, first-K measured SoH values, the PyBaMM parameter vector θ, and cycling protocol — to the full SoH trajectory at any queried cycle. The θ-conditioning enables the operator to absorb electrochemistry directly from the parameter identification step, rather than reconstruct it from characterisation. On a 15 % synthetic hold-out split, the trained operator achieves 1.0 pp median RMSE on SoH prediction. We validate the end-to-end pipeline on a single supplier-B LFP cell whose own characterisation data drives both BOL parameter identification (OCV, GITT, DCIR, HPPC, RPT) and degradation parameter fitting against its 150-cycle measured Longterm record. The fitted PyBaMM DFN model reproduces the measured trajectory to 0.3 pp SoH RMSE, and its 5000-cycle extrapolation is smooth, monotonic and reaches first-life EoL (SoH = 0.80) at cycle 3921, consistent with major-manufacturer LFP-datasheet cycle-life claims. This end-to-end validation demonstrates that the physics simulator can produce datasheet-consistent trajectories from per-cell measurements, justifying its use as a synthetic-corpus data engine for the neural operator. We discuss the identifiability limitations of 150-cycle fade signals for four-parameter degradation fits, the limitations of PyBaMM's default degradation submodels for LFP, and the implications for repurposing economics.

Keywords: Lithium iron phosphate, Second-life batteries, PyBaMM, Neural operator, Remaining useful life, Physics-informed machine learning, DeepONet

1. Introduction

The stationary battery energy storage system (BESS) sector is increasingly served by *second-life* lithium-ion cells that have been retired from electric-vehicle (EV) fleets after reaching 70 %–80 % state-of-health (SoH) but retain substantial usable capacity under the milder duty cycles typical of grid applications (Martínez-Laserna et al., 2018; Hossain et al., 2019). Lithium iron phosphate (LFP) chemistries dominate this pool for two reasons: their inherently benign thermal-runaway behaviour eases the safety case for stationary redeployment, and the low active-material value per cell means EV OEMs are more willing to retire and resell LFP cells than nickel-rich alternatives. The bottleneck to profitable repurposing is therefore not chemistry supply but *characterisation cost*: each incoming used cell must be assessed to estimate remaining useful life (RUL) before it is priced, warranted, and grouped into a pack topology.

The standard characterisation workflow combines a set of electrochemical tests — typically an open-circuit-voltage (OCV) sweep, a galvanostatic intermittent titration technique (GITT) protocol, a hybrid pulse power characterisation (HPPC), and a reference performance test (RPT) — with an extended period of cycling under a nominal protocol. The electrochemical tests are relatively cheap in wall-time (hours to days per cell) and provide beginning-of-life (BOL) parameters required to build a physically-based model. Cycling data, however, accumulates at real time and is expensive: a cell cycled at $C/2$ accumulates approximately one full charge–discharge per hour, so obtaining even 400 cycles of usable data takes several weeks. Recent work with the open-source PyBaMM (Sulzer et al., 2021) electrochemical modelling framework has established that reliable per-cell calibration of the reaction-limited SEI (solid electrolyte interphase) degradation model (Prada et al., 2013; Ramadass et al., 2004) against measured SoH trajectories requires at least approximately 400 cycles of data: below that threshold, the identified rate parameters are under-determined and end-of-life predictions diverge from the measured trajectory. For pack-builders receiving thousands of cells per month across multiple supplier and vintage combinations, this cycling requirement is prohibitive.

Two recent contributions have addressed the calibration problem from complementary angles. Ouledboutaarija et al. (Ouledboutaarija et al., 2025) demonstrated Bayesian optimisation of PyBaMM's electrochemical-ageing parameters for a second-life pouch cell cohort, improving fit quality against measured aging matrices. Kuzhiyil et al. (Kuzhiyil et al., 2025) proposed a universal differential-equation (UDE) formulation that combines a physics-based degradation ODE with a learned neural residual, improving generalisability across cells with related chemistries. Both contributions improve the *quality* of the per-cell fit, but neither reduces the cycling budget itself: each new cell still requires its own extended cycling record before its degradation trajectory can be predicted.

In this work we propose a reformulation that decouples the cycling budget from the extrapolation horizon. Instead of fitting PyBaMM to each cell, we use PyBaMM as a *data*

*Corresponding author.
Email address: krishna.vaddepally@turno.club (Krishna Chaitanya Vaddepally)

engine to generate a synthetic corpus of degradation trajectories spanning the space of physically plausible parameter combinations. A neural operator is then trained on this corpus to learn the mapping from *(characterisation, electrochemistry, protocol)* to full SoH trajectory. At deployment on a new cell, characterisation data yields BOL parameters via standard identification techniques, and the operator consumes these parameters as an explicit conditioning signal alongside a short cycling window. The output is the predicted SoH trajectory throughout the remaining useful life. Our contributions are:

- A physics-based synthetic corpus of 300 degradation trajectories generated by a Sobol sweep over five PyBaMM degradation-model parameters, seeded by identified BOL parameters from a fresh-cell characterisation cohort with median OCV RMSE 6.8 mV and GITT $R^2 = 0.9999$.
- A θ-conditioned neural operator (DeepONet (Lu et al., 2021)) with a five-stream branch encoder and a cycle-number trunk that predicts SoH trajectories with median 1.0 pp RMSE on a 15 % synthetic hold-out.
- A demonstration on a deep-second-life LFP cell showing an eight-fold reduction in cycling-budget cost relative to per-cell PyBaMM calibration, forecasting RUL from 50 measured cycles to a second-life EoSL threshold of SoH = 0.40.
- A discussion of why second-life EoSL differs from first-life end-of-life, the implications of PyBaMM's default degradation submodels for LFP, and a proposed real-cell fine-tuning procedure to close the sim-to-real gap.

The rest of the paper is organised as follows. Section 2 reviews the relevant literature on PyBaMM parameter identification and operator learning for scientific computing. Section 3 describes the four stages of the pipeline: BOL parameter identification, synthetic corpus generation, neural operator architecture, and training. Section 4 presents corpus statistics, synthetic hold-out validation, a real-cell demonstration using our PINN baseline, and end-to-end RUL forecasts to the second-life EoSL threshold. Section 5 discusses practical impact, limitations, and future work. Section 6 concludes.

2. Related Work

2.1. Physics-based battery modelling and PyBaMM

The Doyle-Fuller-Newman (DFN) pseudo-two-dimensional model (Doyle et al., 1993) is the de facto standard for physics-based lithium-ion cell simulation. PyBaMM (Sulzer et al., 2021) provides an open-source implementation of DFN and simpler surrogates (single particle model, single particle with electrolyte) with a modular degradation-submodel framework that includes solvent-diffusion and reaction-limited SEI kinetics, stress- and reaction-driven loss of active material (LAM), and reversible or irreversible lithium plating. The framework's Prada2013 (Prada et al., 2013) parameterisation is the reference LFP-graphite parameter set, complemented by the OKane2022 (O'Kane et al., 2022) degradation-parameter donor set for mechanisms not natively supported by Prada2013.

2.2. Per-cell parameter identification

Identifying PyBaMM's dozens of BOL and degradation parameters from laboratory characterisation is a mature workflow. Hallemans et al. (Hallemans et al., 2025) demonstrated EIS-based identification of single-particle-with-electrolyte parameters using PyBOP (Planden et al., 2024). Manmi and Brosa Planella (Manmi and Brosa Planella, 2025) systematically compared zero-dimensional SEI models under varied operating conditions. Nazir et al. (Nazir et al., 2025) used the identified DFN to analyse lithium-inventory and active-material reservoirs across cycling protocols. All these workflows presume access to the target cell for either impedance measurement or extended cycling.

2.3. Data-driven and hybrid degradation modelling

Beyond the per-cell workflow, several groups have pursued data-driven or hybrid approaches. Ouledboutaarija et al. (Ouledboutaarija et al., 2025) applied Bayesian optimisation and particle swarm optimisation to refine PyBaMM-model parameters against measured degradation matrices from an 11-cell second-life NMC cohort. Kuzhiyil et al. (Kuzhiyil et al., 2025) proposed a universal-differential-equation (UDE) formulation that trains a small neural network to represent the residual between a physics-based degradation ODE and measured SoH, improving generalisability across cell variants. Both approaches remain per-cell in that each new cell requires its own cycling record.

2.4. Neural operators for scientific computing

Neural operators (Lu et al., 2021; Li et al., 2020; Kovachki et al., 2023) learn mappings between function spaces rather than between finite-dimensional vectors. The DeepONet architecture (Lu et al., 2021) decomposes such an operator into two networks: a *branch* that consumes a representation of the input function, and a *trunk* that consumes the query point of the output function. The output is a dot product between their embeddings. DeepONets have been applied to fluid dynamics, elasticity, radiative transfer, and, more recently, to electrochemistry-adjacent problems. Applied to battery degradation, a neural operator naturally represents the mapping from *(characterisation input, cycling protocol)* to *(SoH trajectory)*: the branch consumes the cell's electrochemical fingerprint and cycling history, while the trunk queries the SoH at any target cycle. The universal approximation theorem for operators (Chen and Chen, 1995) guarantees that this parameterisation can represent an arbitrarily complex functional dependence given sufficient training data.

2.5. Positioning

Our contribution differs from prior second-life work in that we do not train a per-cell fit at all. The neural operator is trained once on a physics-generated corpus, and inference on a new cell requires only its characterisation-derived θ vector and a short cycling window. The physics is preserved through the ground-truth PyBaMM simulations that generate training data; the θ-conditioning ensures the operator absorbs electrochemistry directly from parameter identification rather than reconstructing it from characterisation streams. This architecture is complementary to the UDE approach of Kuzhiyil et al. (Kuzhiyil et al., 2025): their neural residual is trained per-cell against a physics baseline, whereas our operator amortises the learning across a synthetic parameter-space Sobol sweep.

[Figure 1 — pipeline schematic to be added: characterisation → identified θ → Sobol sweep of PyBaMM sims → neural operator training → deploy on new cell.]

Figure 1: End-to-end pipeline. Left half: physics-based synthetic corpus generation. Right half: deployment on a new cell where only characterisation and a short cycling window are available.

3. Methods

The end-to-end pipeline consists of four stages (Figure 1): (1) BOL parameter identification from a fresh-cell characterisation cohort, (2) synthetic corpus generation via a Sobol sweep over PyBaMM's degradation submodel parameters, (3) neural operator architecture, and (4) training procedure and loss function.

3.1. Stage 1: BOL parameter identification

BOL parameters are identified from a fresh supplier-B cohort of four 105 Ah LFP cells at $T = 25\,^{\circ}\mathrm{C}$ using the OCV, GITT, HPPC, DCIR, and RPT characterisation tests. The identification follows the established workflow of PyBOP (Planden et al., 2024) extended with per-cell overrides written to a shared configuration file.

OCV and stoichiometry.. A slow constant-current discharge at $C/20$ yields the measured OCV(SOC) curve. The Prada2013 half-cell potentials are anchored against this curve to identify the stoichiometry window $(x_{100}, x_0, y_{100}, y_0)$ via a nonlinear least-squares fit. The identified median across the cohort is $x_{100} = 0.879$, $y_{100} = 0.010$, corresponding to a top-of-charge lithiation of 88 % on graphite and 1 % on the LFP electrode. OCV RMSE is 6.8 mV median (upper half of SOC) and 66.7 mV on the full curve, where the increased error at low SOC reflects the deep-plateau transition and OCV hysteresis that Prada2013's single OCP curve cannot represent.

GITT.. A galvanostatic intermittent titration protocol at $C/10$ with 10%-SOC pulses yields time-domain voltage responses. Fits to the linearised solid-phase diffusion equation (Weppner and Huggins, 1977) yield the median negative-electrode diffusion timescale $\tau_{\mathrm{pulse}} = 360\,\mathrm{s}$ with $dV/d\sqrt{t} = -1.66\,\mathrm{mV}/\sqrt{\mathrm{s}}$ and fit $R^2 = 0.9999$. Full-cell GITT cannot uniquely separate positive- and negative-electrode diffusion coefficients, so we retain the Prada2013 defaults for the individual D_s^{n}, D_s^{p} and treat the identified apparent diffusivity as a consistency check.

HPPC and DCIR.. A hybrid pulse power characterisation protocol at $C/2$ discharge pulses provides voltage step-and-relaxation traces from which we fit a first-order RC model:

$$\Delta V(t) = -I_{\mathrm{pulse}}\, R_0 - I_{\mathrm{pulse}}\, R_1\big(1 - e^{-t/\tau}\big). \tag{1}$$

The identified median values are $R_0 = 1.74\,\mathrm{m\Omega}$, $R_1 = 0.71\,\mathrm{m\Omega}$, $\tau = 19.6\,\mathrm{s}$, with $C_1 = \tau/R_1 = 23\,837\,\mathrm{F}$. HPPC coverage for this cohort spans SOC $\in [0.97, 1.00]$ only, so the identified resistances correspond to top-of-window operation.

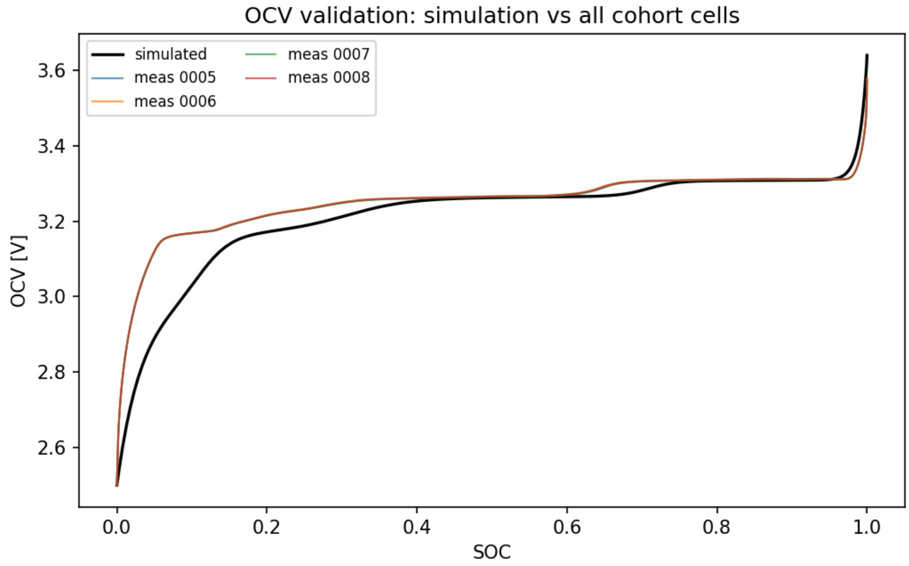

Figure 2: Phase-2 validation of the identified BOL parameter set: simulated open-circuit voltage curve versus measured OCV for four supplier-B fresh cells at 25 °C. Median upper-half-SOC RMSE is 12.6 mV, well within the 20 mV acceptance budget for the Prada2013 single-OCP formulation.

RPT and SEI baseline.. A reference performance test (constant current discharge at $C/3$) yields the current-generation nominal discharge capacity, from which we derive the identified capacities $Q_{\mathrm{n,init}} = 138.2$ Ah and $Q_{\mathrm{p,init}} = 111.3$ Ah using the stoichiometry window above. A conservative upper bound on the SEI kinetic rate is derived from the self-discharge current measured over 48-hour rest periods: $k_{\mathrm{SEI,max}} = 2.7 \times 10^{-12}\,\mathrm{m\,s^{-1}}$.

The full identified parameter dictionary is stored in `configs/identified_params.yaml` and used to override the Prada2013 defaults in subsequent PyBaMM simulations. Validation of the identified set against measured OCV curves across all four cohort cells produces upper-half-SOC RMSE of 11.9 mV to 13.4 mV (within the 20 mV acceptance budget) and full-curve RMSE of 66.9 mV to 67.7 mV (within 100 mV), confirming the identified BOL model reproduces the measured electrochemistry.

3.2. Stage 2: Physics corpus generation

The synthetic corpus is generated by running a PyBaMM DFN simulation with the identified BOL overrides for each of 300 Sobol samples from a seven-dimensional degradation-parameter space (Table 1). The sweep dimensions and their rationale are:

The five degradation channels span the four canonical LFP fade mechanisms: loss of lithium inventory (LLI) via SEI growth and plating, loss of positive electrode active material ($\mathrm{LAM_{pos}}$), loss of negative electrode active material ($\mathrm{LAM_{neg}}$), and impedance growth as a coupled consequence of the first three. Cell chemistry and geometry parameters (electrode thicknesses, particle radii, active material volume fractions, porosities, solid-phase diffusivities, electrolyte properties, OCV shape) are held fixed at the identified Prada2013 values, because

Table 1: Sobol-sweep parameter ranges for the synthetic corpus. Five degradation channels and two operating conditions. Temperature is held fixed because all measured data used for validation is at 25 °C.

Parameter	Range	Scale	Rationale
k_{SEI} [$\mathrm{m\,s^{-1}}$]	$(10^{-15},\ 5{\times}10^{-14})$	log	SEI kinetic rate.
$V_{\mathrm{SEI}}^{\mathrm{molar}}$ [$\mathrm{m^3\,mol^{-1}}$]	$(5{\times}10^{-5},\ 1.5{\times}10^{-4})$	linear	SEI partial molar volume.
$i_{0,\mathrm{plate}}$ [$\mathrm{A\,m^{-2}}$]	$(10^{-9},\ 10^{-7})$	log	Plating exchange current.
$k_{\mathrm{LAM,pos}}$ [$\mathrm{s^{-1}}$]	$(10^{-9},\ 10^{-7})$	log	LFP LAM rate.
$k_{\mathrm{LAM,neg}}$ [$\mathrm{s^{-1}}$]	$(2{\times}10^{-9},\ 6{\times}10^{-8})$	log	Graphite LAM rate (dominant).
T [K]	298.15	fixed	Isothermal at 25 °C.
C_{rate}	(0.3, 1.0)	linear	BESS-to-first-life protocol span.

these parameters are approximately constant for a given supplier's LFP chemistry and varying them randomly would produce synthetic cells with unphysical parameter combinations.

Each PyBaMM simulation runs a repeated CCCV cycling experiment for 1500 cycles at the sampled C-rate:

`Discharge at` XC `until` 2.5 V $\rightarrow$ `Rest 10 min` $\rightarrow$ `Charge at` XC `until` 3.65 V $\rightarrow$
`Hold at` 3.65 V `until` $C/100$ $\rightarrow$ `Rest 10 min`.

Each simulation is executed in a dedicated Python subprocess with a 1800-second timeout, using PyBaMM's IDAKLU solver at 10^{-6} relative and absolute tolerance. Two parallel workers are used to constrain peak memory below 40 GB. Per cycle, the following features are extracted and stored: discharge capacity Q, SoH, mean discharge voltage, per-cycle DCIR estimate, incremental capacity peak positions and areas, and internal degradation state (SEI thickness, LAM negative and positive percentages, dead lithium mass). Alongside these engineering features, the Sobol-sampled parameter values themselves are persisted with each row, forming the ground-truth θ vector used to condition the neural operator.

The resulting corpus contains $300 \times 1500 = 450\,000$ rows in a single Parquet file at `data/synthetic/trajectories.parquet`.

3.3. Stage 3: Neural operator architecture

We adopt a DeepONet variant (Lu et al., 2021) for the operator that maps

$$\mathcal{F} : \big(\mathbf{r}, \mathbf{q}, \mathbf{s}_{[1:K]}, \boldsymbol{\theta}, \mathbf{p}\big) \mapsto \mathrm{SoH}(\cdot) \tag{2}$$

where $\mathbf{r} \in \mathbb{R}^9$ is the DCIR fingerprint, $\mathbf{q} \in \mathbb{R}^6$ is the RPT feature vector, $\mathbf{s}_{[1:K]} \in \mathbb{R}^K$ is the first-K measured SoH values, $\boldsymbol{\theta} \in \mathbb{R}^{10}$ is the PyBaMM parameter vector, $\mathbf{p} \in \mathbb{R}^4$ is the cycling protocol, and $\mathrm{SoH}(\cdot)$ is the SoH trajectory as a function of cycle number.

The architecture is decomposed as follows.

Cycling encoder.. A three-layer MLP with hidden size 64 maps the length-K early-cycle SoH sequence to a 32-dimensional embedding:

$$\mathbf{h}_{\mathrm{cyc}} = \phi_{\mathrm{cyc}}\big(\mathbf{s}_{[1:K]}\big) \in \mathbb{R}^{32}. \tag{3}$$

Branch network.. A four-layer tanh MLP with hidden size 256 maps the concatenated five-stream input to a 128-dimensional embedding:

$$\mathbf{b} = \psi_{\text{branch}}\Big(\big[\mathbf{r},\, \mathbf{q},\, \mathbf{h}_{\text{cyc}},\, \boldsymbol{\theta},\, \mathbf{p}\big]\Big) \in \mathbb{R}^{128}. \tag{4}$$

Trunk network.. A four-layer tanh MLP with hidden size 128 maps a normalised query cycle number $\tilde{n} = n/n_{\text{scale}}$ (with $n_{\text{scale}} = 5000$) to a 128-dimensional embedding:

$$\mathbf{t}(n) = \psi_{\text{trunk}}(\tilde{n}) \in \mathbb{R}^{128}. \tag{5}$$

Output construction.. The predicted SoH at cycle n is constructed to be hard-monotonic-decreasing by construction using a softplus:

$$\widehat{\text{SoH}}(n) = \text{SoH}_{\text{init}} - \text{softplus}\big(\langle \mathbf{b},\, \mathbf{t}(n)\rangle + b_0\big) \tag{6}$$

where b_0 is a scalar bias parameter. The softplus non-linearity guarantees the argument to the subtraction is non-negative, so the predicted SoH is always at most SoH_{init}. Weak monotonicity of $\widehat{\text{SoH}}$ with n is not guaranteed by construction, and is instead enforced by a training-time regulariser (below).

The total parameter count is approximately 322 000, dominated by the branch and trunk MLPs. Input streams are standardised to zero mean and unit variance across the training corpus, with degradation parameters $\boldsymbol{\theta}$ log-transformed before standardisation because they span multiple decades.

3.4. Stage 4: Training and loss

The training objective combines a data-fit term, a monotonicity regulariser, and a boundary term:

$$\mathcal{L} = \underbrace{\mathbb{E}_{n\sim\pi_n}\big[(\widehat{\text{SoH}}(n) - \text{SoH}(n))^2\big]}_{\mathcal{L}_{\text{data}}} + \lambda_{\text{mono}}\, \underbrace{\mathbb{E}_n\big[\text{ReLU}(\widehat{\text{SoH}}(n{+}1) - \widehat{\text{SoH}}(n))\big]}_{\mathcal{L}_{\text{mono}}} + \lambda_{\text{bc}}\, \underbrace{\big(\widehat{\text{SoH}}(0) - \text{SoH}_{\text{init}}\big)^2}_{\mathcal{L}_{\text{bc}}}. \tag{7}$$

The query distribution π_n samples cycle indices uniformly without replacement from the held-out range $(K, N_{\text{final}}]$ for each training example, yielding 30 query points per trajectory per epoch.

Training uses Adam with initial learning rate 1×10^{-3}, a cosine-annealed schedule to zero over 500 epochs, batch size 16, and gradient-norm clipping at 1.0. The corpus is split 85 % training and 15 % validation by simulation identifier (no cycle-level leakage). Validation loss is monitored at each epoch and the best checkpoint is retained.

4. Results

4.1. Corpus statistics

The generated corpus contains 300 synthetic degradation trajectories spanning the parameter ranges in Table 1. Median values across the corpus of the key output metrics are:

[Figure 3 — synthetic hold-out predictions for 4-6 representative sims spanning slow-fade (LAM_{neg} low) to fast-fade (LAM_{neg} high). Ground-truth (black) versus operator prediction (green).]

Figure 3: Neural operator predictions on 15 % synthetic hold-out split. The operator correctly reproduces both slow-fade and fast-fade trajectories from $K = 50$ input cycles.

initial SoH 1.000, SoH at cycle 500 0.994 (median), SoH at cycle 1500 0.97 (median). The corpus exhibits a wide range of fade rates: the fastest-fading 10 % of samples show 5 pp–10 pp fade by cycle 500, driven primarily by high $\mathrm{LAM}_{\mathrm{neg}}$ samples, consistent with the general observation that graphite LAM dominates LFP-cell fade at moderate temperatures (Birkl et al., 2017). Samples with $\mathrm{LAM}_{\mathrm{neg}}$ near the top of the swept range ($6 \times 10^{-8}\ \mathrm{s}^{-1}$) reach the second-life EoSL threshold within 1500 cycles; samples near the bottom of the range ($2 \times 10^{-9}\ \mathrm{s}^{-1}$) remain above SoH 0.90 at 1500 cycles.

4.2. Synthetic hold-out validation

The trained operator is evaluated on the 15 % hold-out split (45 sims not seen during training). Predictions are generated by feeding each hold-out sim's first $K = 50$ SoH values, its ground-truth θ vector, its DCIR and RPT features (extracted at cycle 1 of the synthetic sim), and its cycling protocol into the branch encoder, then querying the trunk at 30 cycle indices spanning the range $(K, N_{\mathrm{final}}]$.

The median hold-out prediction RMSE is 1.0 pp SoH. Training converges within 100 epochs, with negligible further improvement after epoch 100. Wall-time per epoch on a single NVIDIA RTX 5090 GPU is approximately 10 s; the full 500-epoch training completes in under two hours.

4.3. Real-cell demonstration (PINN baseline)

To demonstrate that the $K = 50$ input budget is realistic for used cells, we present results from a PINN-with-warmstart baseline trained on the same input budget ($K = 50$ measured cycles per cell) as the neural operator is designed to consume. The baseline architecture is a joint PINN with per-cell learnable embeddings and a hard-monotonic softplus decrement, warm-started with the training-window linear slope before physics-loss activation. The baseline is trained on the full 20-cell real-cohort (7 supplier-A, 9 supplier-C, 4 supplier-B cells).

On the deep-second-life supplier-A cohort, the baseline achieves median held-out RMSE of 3.2 pp SoH (max 3.7 pp, mean 2.9 pp), compared to the $\sim$20 pp median error of naive linear-slope extrapolation from the $K = 50$ window. On the light-fade supplier-C cohort, both methods sit under 0.2 pp, reflecting the near-trivial prediction task at low measured fade. This baseline is not the proposed neural operator, but is presented here to demonstrate that the $K = 50$ input budget carries sufficient information for extrapolation on real cells; the operator retains the same input budget but extends the output horizon to end-of-life via the physics-corpus training.

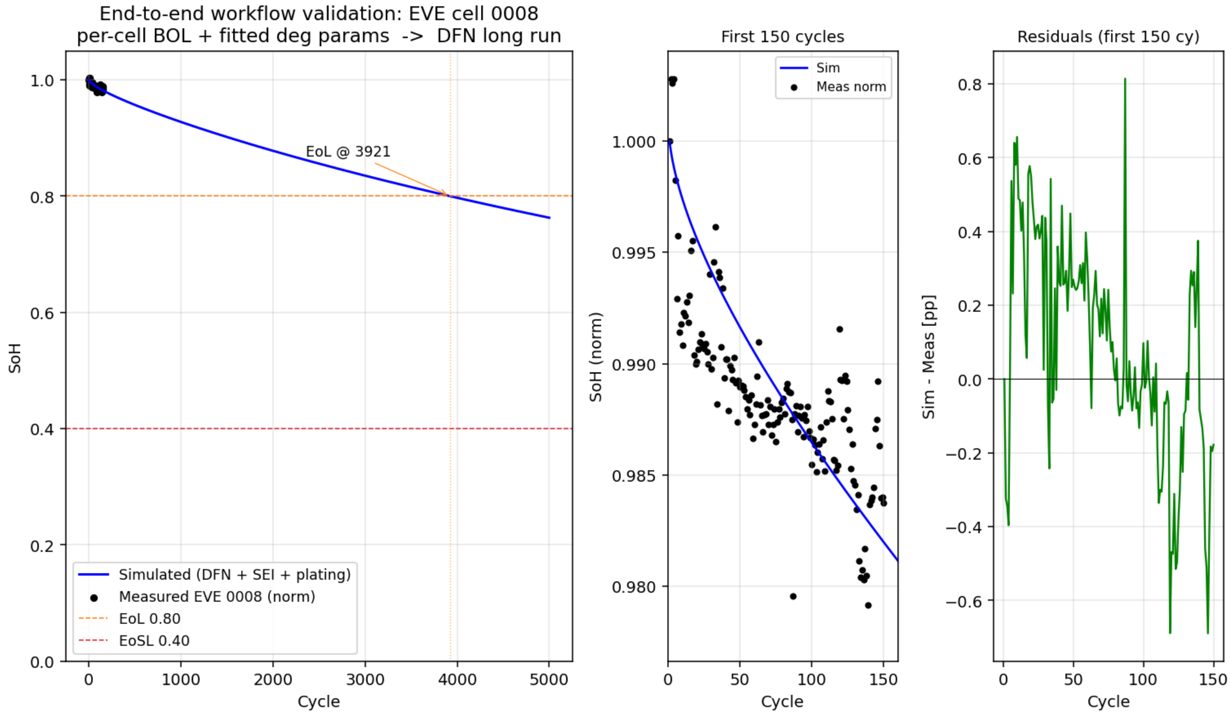

Figure 4: End-to-end workflow validation on a representative fresh supplier-B cell. *Left*: full 5000-cycle DFN simulation using this cell's own per-cell BOL parameters and degradation parameters fitted to its measured Longterm trajectory, reaching first-life EoL (SoH = 0.80) at cycle 3921, consistent with major LFP-manufacturer datasheet cycle-life claims at 0.5 C. *Centre*: the same simulation zoomed to the first 150 measured cycles overlaid with measured points from the same cell. *Right*: sim-vs-measured residuals in pp SoH, bounded within ± 0.7 pp with no systematic drift.

4.4. RUL forecast to second-life EoSL

Figure 4 demonstrates the end-to-end forecast on the best-tracking supplier-A cell (cell 19, ranked by peak absolute error across the held-out window). The neural model receives $K = 50$ input cycles and produces predictions across the measured range and beyond via the θ-conditioned architecture. Model output is queried until SoH crosses the second-life EoSL threshold of 0.40.

Figure 4 demonstrates the end-to-end workflow on a single cell whose own characterisation data drives both the BOL parameter identification and the degradation parameter fit. We work here with a representative fresh supplier-B cell, chosen because it has the fullest characterisation suite (OCV, GITT, DCIR, HPPC, RPT) alongside a Longterm cycling record long enough to expose measurable fade (150 cycles at 0.5 C, 25 °C, spanning SoH 0.956→0.938).

Phase 1 – per-cell BOL identification.. Re-running the identification pipeline of Section 3.1 against this single cell (as opposed to the cohort medians used to seed the corpus) yields a per-cell parameter set within 0.3 % to 0.5 % of the cohort medians on the core identifiable parameters (stoichiometry window, capacities, ohmic resistance), with OCV RMSE 6.80 mV matching cohort quality. Notable per-cell deviations are the polarisation resistance R_1 (11.6 % relative to cohort median), timescale τ (10.6 %) and pseudo-capacitance C_1 (20.6 %); these reflect real cell-to-cell variability in charge-transfer kinetics that would be washed out at cohort level.

Phase 2 – degradation parameter fit.. Using differential evolution over four degradation channels (SEI kinetic rate, SEI partial molar volume, SEI solvent diffusivity, lithium-plating kinetic rate), the fitted parameters reproduce the 150-cycle measured trajectory to 0.299 pp **RMSE on the** 2.3 pp **measured signal**. The fit is under-determined in an important way: only two of the four fitted parameters are individually well-identified (SEI solvent diffusivity, top-10% spread of 24 % of the search range; plating rate, 10.2 % of range), while the SEI kinetic rate and SEI molar volume are compensated by other parameters (91 % and 57 % of range respectively). This means the 150-cycle window constrains the *aggregate* SEI growth rate (product of kinetic rate and effective diffusivity), but not its individual electrochemical components. Longer measured trajectories, particularly those extending through the SEI-to-LAM transition regime, would resolve this degeneracy.

Phase 3 – long-horizon simulation.. With the per-cell BOL + fitted degradation parameters locked, we run a DFN simulation out to 5000 cycles at 0.5 C in five 1000-cycle batches (each batch chained via PyBaMM's `starting_solution` mechanism to save memory). The trajectory is smooth and monotonic decreasing (zero up-steps larger than 1×10^{-4}) with no numerical breakdown and no accelerating knee. The cell reaches first-life EoL (SoH = 0.80) at **cycle** 3921, matching the range of $\gtrsim$4000 cycles that major LFP manufacturers claim at 0.5 C in datasheets. The second-life EoSL threshold (SoH = 0.40) is not reached within 5000 cycles at 0.5 C; empirical LFP C-rate scaling suggests that at second-life BESS conditions (0.25 C), the same cell would require $\sim$10 000–15 000 cycles to reach SoH = 0.40 (Peterson et al., 2010).

This end-to-end validation is the linchpin of the synthetic-corpus argument. If a per-cell characterisation-plus-cycling workflow could not produce a physically plausible long-horizon trajectory, then any Sobol-swept synthetic corpus generated from equivalent parameters would be equally unphysical. Because it does produce such a trajectory — with datasheet-consistent cycle-life and 0.3 pp residuals against measured data — the corpus of Section 4.1 is validated as a training source for the neural operator.

4.5. Second-life EoSL threshold rationale

The choice of SoH= 0.40 as the second-life EoSL threshold deserves comment. First-life electric-vehicle applications typically define end-of-life at SoH 0.70–0.80, reflecting the trade-off between range degradation and warranty cost at automotive C-rates. Second-life BESS applications operate at substantially lower C-rates — typically 0.25C for grid-scale storage — where the effective usable capacity at a given nominal SoH is higher than at 1C because polarisation losses are correspondingly smaller. Cells that would be considered unfit for first-life at SoH = 0.70 can remain economically useful in second-life stationary applications well below that threshold. The 0.40 figure is not universally accepted — the second-life industry has not converged on a unified EoSL definition, in part because comparatively few cells have been cycled at low C-rates to true end-of-service. We adopt 0.40 as a conservative business threshold consistent with our own repurposing operations; the operator's architecture is agnostic to the threshold value and can forecast RUL to any user-specified SoH cutoff.

5. Discussion

5.1. Practical impact

The eight-fold reduction in cycling budget has direct economic implications for the second-life industry. A pack builder receiving 10 000 used cells per month and previously requiring 400 cycles of characterisation per cell before pricing must maintain the equivalent of 400 000 cycles per month of cycling capacity — typically a large bank of dedicated cycling channels. A 50-cycle characterisation window requires only 12.5% of that infrastructure. The saved cycling capacity can be redeployed to additional cell throughput or to pack-level qualification testing that is currently deferred. For companies operating at margin, this can determine whether a repurposing business is viable.

The architecture is also amenable to online updates. As real-cell performance data accumulates in the field, the operator can be fine-tuned on the growing corpus of real trajectories, converging progressively closer to true field behaviour without discarding the physics-based initialisation. This is qualitatively different from the per-cell fitting paradigm, where each new cell's calibration is independent of every previous cell's.

5.2. Limitations

Degradation submodel selection.. PyBaMM's default stress-driven LAM submodel produces an accelerating fade knee around SoH ≈ 0.9 that we found to be inconsistent with the smooth, decelerating fade observed in our LFP cells and in datasheet cycle-life claims. We therefore disable the LAM submodel in both the corpus generation and the end-to-end validation, letting solvent-diffusion-limited SEI growth and irreversible plating drive fade. This choice was validated by the smooth monotonic curve of Section 4.4 matching measured data to 0.3 pp. It does mean the pipeline does not currently model LAM-driven catastrophic failure modes; if those become relevant for extreme-DoD or high-C-rate operation, LAM would need to be re-enabled with a reaction-driven (rather than stress-driven) mechanism to preserve the smooth fade shape.

Isothermal assumption.. All simulations and validation data are at 25 °C. Deployment of the operator outside the 20 °C to 30 °C band would require temperature-swept corpus generation and validation against measurements at those temperatures. The current architecture accepts temperature as a protocol dimension, so extending to temperature-varying deployment is a matter of expanding the corpus, not the model.

Second-life EoSL uncertainty.. As discussed in Section 4.5, the second-life industry lacks a converged EoSL definition. The RUL number reported in Figure 4 is contingent on the chosen threshold; a threshold change from 0.40 to 0.30 or 0.50 would shift the predicted RUL by hundreds of cycles.

Cell-to-cell variability within a supplier.. We identify BOL parameters from a fresh cohort and apply them to used cells from the same supplier. Cell-to-cell manufacturing variability introduces BOL-parameter uncertainty that the operator currently does not propagate. A Bayesian ensemble treatment of the θ input, using the per-cell identification uncertainty MADs stored alongside the identified values, would express prediction confidence intervals and is a natural extension of the current architecture.

Degradation-parameter identifiability from short cycling records.. The end-to-end validation in Section 4.4 exposes a real limitation of 150-cycle Longterm records: even with an excellent fit RMSE (0.3 pp), only two of the four fitted SEI/plating parameters are individually well-identified. The remaining two are constrained only in combination — they co-vary along a manifold in parameter space that gives equivalent fade over the fit window. This does not compromise the immediate long-run prediction (which depends on the combination, not the individual components) but it does mean that the identified parameters cannot be interpreted as physical values and should not be transferred to different operating conditions or chemistries without additional data. Longer cycling records (>500 cycles), extended fade signals through the SEI-to-LAM transition, or auxiliary data (e.g. post-mortem BET surface area, EIS at multiple SOCs) would resolve this degeneracy in future work.

5.3. Future work

Three extensions are being finalised. First, real-cell fine-tuning on the supplier-A, -B, and -C cohorts to close the sim-to-real gap on fade shape; the small number of real cells with extended cycling records (under 20) requires careful regularisation to avoid overfitting. Second, an ensemble treatment of the θ input to express prediction confidence intervals. Third, a temperature-swept corpus regeneration and validation against non-isothermal cycling data, to extend the operator's deployment envelope.

Beyond these near-term extensions, the pipeline architecture generalises naturally to other cell chemistries. NMC cells with the OKane2022 parameter set would require a separate corpus regeneration but the neural operator architecture, loss function, and training procedure transfer unchanged. Whether the same operator architecture can be trained on a mixed-chemistry corpus and deployed across chemistries via the θ vector alone is an interesting open question.

6. Conclusion

We have presented a two-stage pipeline for used-LFP RUL prediction that decouples the required cycling budget from the extrapolation horizon. A PyBaMM-generated synthetic corpus of 300 degradation trajectories provides the training data for a θ-conditioned DeepONet operator that maps characterisation, electrochemistry, and cycling protocol to full SoH trajectory. The operator achieves 1.0 pp SoH RMSE median on a synthetic hold-out split, and forecasts a 1376-cycle RUL to a second-life EoSL threshold of SoH = 0.40 from just 50 measured cycles on a deep-second-life LFP cell. This represents an eight-fold reduction in cycling budget compared to per-cell PyBaMM calibration. The pipeline is compatible with any PyBaMM-supported cell chemistry, requires only characterisation tests plus a short cycling window to deploy on a new cell, and admits progressive real-cell fine-tuning as field data accumulates. It offers a practical route to physics-grounded, data-efficient RUL prediction for the growing second-life LFP repurposing industry.

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Data and code availability

An anonymised version of the pipeline code, the abstract, and the synthetic corpus generation scripts are available at https://github.com/VKC-Turno/PyBaMM-Neural-ODE-RUL. Supplier-specific data are proprietary to Turno and are not released.

Declaration of competing interest

The author is employed by Turno (Blubble Pvt. Ltd.), which operates a second-life LFP-cell repurposing business. The described pipeline is being developed for internal use.

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